

Álvaro López Álvarez

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LinkedIn, GitHub, Portfolio

Software Engineer who moves comfortably across the stack, choosing the language and tools each problem calls for rather than defaulting to what's familiar: an LLM agent in Python/LangGraph, event-driven microservices in Java and Kafka, a distributed Function-as-a-Service platform in Go, a language interpreter and an HTTP server built from raw sockets. Drawn to owning a problem end to end, from first principles to production, and to cutting complexity that doesn't earn its keep.

Core Competencies

- **Backend & Systems Programming**: Golang, TypeScript/JavaScript, Java, Python, Kotlin, C, SQL
- **Distributed Architecture**: Docker, event-driven microservices (Kafka, NATS), API gateways (Apache APISIX), horizontal scaling
- **Full-Stack Ownership**: React, PostgreSQL, REST APIs, OAuth 2.0/JWT
- **LLM/Agentic AI Engineering**: LangChain, LangGraph, RAG pipelines, safety-conscious agent design
- **Deep Systems Engineering**: hand-built HTTP/1.1 server on raw TCP sockets, tree-walking language interpreter (lexer, parser, AST, evaluator)

Professional Experience

- **Instituto Tecnológico de Informática (ITI)** — **Valencia, España**
  R&D Developer — *Feb 2026–Jun 2026*
  - Took an open-ended problem, natural-language queries against a sensitive data-governance schema, to a working system: a LangGraph agent with schema selection, structured SQL generation, execution validation, and retry-on-error
  - Implemented LLM-based improvements to a search engine end to end
  - Applied safety-first design discipline throughout, since the agent only ever produced validated, read-only SQL against production data
- **Instituto Tecnológico de Informática (ITI)** — **Valencia, España**
  Full-Stack Developer Intern — *Feb 2024–Jun 2024*
  - Owned frontend feature delivery in React and TypeScript for the organization's internal personnel-management application, improving usability
  - Contributed across the stack, integrating frontend and backend services

Selected Projects

Self-initiated, built outside coursework and internships to go deeper on systems fundamentals.

- **FaaS with Golang**: Distributed Function-as-a-Service platform; Go workers execute arbitrary Docker-image functions, NATS JetStream work-queues for reliable task distribution, Apache APISIX gateway for JWT auth and rate limiting, horizontal scaling via Docker Compose replicas.
- **Kafka + Spring Boot Ticketing Platform**: Event-driven microservices (reservation, payment, notification, audit); outbox pattern, dead-letter topics, idempotent processing, Testcontainers integration tests.
- **Monkey Interpreter & HTTP Server**: Hand-written lexer, Pratt parser, AST, and evaluator for a custom language, plus an HTTP/1.1 server built directly on raw TCP sockets with no external framework, including chunked transfer encoding and a reverse proxy.

Education

- **Polytechnic University of Valencia (UPV)** **Valencia, España**
  Master's Degree in Computer Engineering (MUIINF) *2024–2026*
  In progress (expected Sep 2026). Topics: distributed systems, high-performance computing, cybersecurity, embedded systems.
- **Polytechnic University of Valencia (UPV)** **Valencia, España**
  Grado (BSc) in Ingeniería Informática, Software Engineering track *2020–2024*
  GPA: 8.88/10. Thesis: "ValenMove - Unified Public Transport App" (grade 9/10).

Languages

- Spanish (native), Catalan (native), English (B2, confident conversational).

References

- Available upon request.